



Syllabus: Limits, Continuity, and Differentiability, Maxima and minima, Mean value theorem, Integration.

Mark Distribution in Previous GATE

Year	2025 - 1	2025 - 2	2024 - 1	2024 - 2	2023	2022	2021 - 1	2021 - 2	Minimum	Average	Maximum
1 Mark Count	1	1	1	1	2	1	1	1	1	1.13	2
2 Marks Count	0	0	0	0	0	0	0	0	0	0	0
Total Marks	1	1	1	1	2	1	1	1	1	1.13	2

5.1 Continuity (9)

5.1.1 Continuity: GATE CSE 1996 | Question: 3



Let f be a function defined by

$$f(x) = \begin{cases} x^2 & \text{for } x \leq 1 \\ ax^2 + bx + c & \text{for } 1 < x \leq 2 \\ x + d & \text{for } x > 2 \end{cases}$$

Find the values for the constants a , b , c and d so that f is continuous and differentiable everywhere on the real line.

gate1996 calculus continuity differentiation normal descriptive

Answer key

5.1.2 Continuity: GATE CSE 1998 | Question: 1.4



Consider the function $y = |x|$ in the interval $[-1, 1]$. In this interval, the function is

- A. continuous and differentiable
- B. continuous but not differentiable
- C. differentiable but not continuous
- D. neither continuous nor differentiable

gate1998 calculus continuity differentiation easy

Answer key

5.1.3 Continuity: GATE CSE 2007 | Question: 1



Consider the following two statements about the function $f(x) = |x|$:

- P. $f(x)$ is continuous for all real values of x .
- Q. $f(x)$ is differentiable for all real values of x .

Which of the following is **TRUE**?

- A. P is true and Q is false.
- B. P is false and Q is true.
- C. Both P and Q are true.
- D. Both P and Q are false.

gatecse-2007 calculus continuity differentiation easy

Answer key

5.1.4 Continuity: GATE CSE 2013 | Question: 22



Which one of the following functions is continuous at $x = 3$?

- A. $f(x) = \begin{cases} 2, & \text{if } x = 3 \\ x - 1 & \text{if } x > 3 \\ \frac{x+3}{3} & \text{if } x < 3 \end{cases}$
- B. $f(x) = \begin{cases} 4, & \text{if } x = 3 \\ 8 - x & \text{if } x \neq 3 \end{cases}$
- C. $f(x) = \begin{cases} x + 3, & \text{if } x \leq 3 \\ x - 4 & \text{if } x > 3 \end{cases}$
- D. $f(x) = \begin{cases} \frac{1}{x^3 - 27} & \text{if } x \neq 3 \end{cases}$

gatecse-2013 calculus continuity normal

Answer key

5.1.5 Continuity: GATE CSE 2014 Set 1 | Question: 47



A function $f(x)$ is continuous in the interval $[0, 2]$. It is known that $f(0) = f(2) = -1$ and $f(1) = 1$. Which one of the following statements must be true?

- A. There exists a y in the interval $(0, 1)$ such that $f(y) = f(y + 1)$
- B. For every y in the interval $(0, 1)$, $f(y) = f(2 - y)$
- C. The maximum value of the function in the interval $(0, 2)$ is 1
- D. There exists a y in the interval $(0, 1)$ such that $f(y) = -f(2 - y)$

gatecse-2014-set1 calculus continuity normal

Answer key

5.1.6 Continuity: GATE CSE 2015 Set 2 | Question: 26



Let $f(x) = x^{-\left(\frac{1}{3}\right)}$ and A denote the area of region bounded by $f(x)$ and the X-axis, when x varies from -1 to 1 . Which of the following statements is/are TRUE?

- I. f is continuous in $[-1, 1]$
- II. f is not bounded in $[-1, 1]$
- III. A is nonzero and finite

- A. II only
- B. III only
- C. II and III only
- D. I, II and III

gatecse-2015-set2 calculus continuity functions normal

Answer key

5.1.7 Continuity: GATE CSE 2021 Set 2 | Question: 25



Suppose that $f: \mathbb{R} \rightarrow \mathbb{R}$ is a continuous function on the interval $[-3, 3]$ and a differentiable function in the interval $(-3, 3)$ such that for every x in the interval, $f'(x) \leq 2$. If $f(-3) = 7$, then $f(3)$ is at most _____

gatecse-2021-set2 numerical-answers calculus continuity one-mark

Answer key

5.1.8 Continuity: GATE DS&AI 2024 | Question: 27



Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function. Note: $\mathbb{R}$ denotes the set of real numbers.

$$f(x) = \begin{cases} -x, & \text{if } x < -2 \\ ax^2 + bx + c, & \text{if } x \in [-2, 2] \\ x, & \text{if } x > 2 \end{cases}$$

Which ONE of the following choices gives the values of a, b, c that make the function f continuous and differentiable?

- A. $a = \frac{1}{4}, b = 0, c = 1$
- B. $a = \frac{1}{2}, b = 0, c = 0$
- C. $a = 0, b = 0, c = 0$
- D. $a = 1, b = 1, c = -4$

gate-ds-ai-2024 calculus continuity differentiation two-marks

Answer key

5.1.9 Continuity: GATE2010 ME



The function $y = |2 - 3x|$

- A. is continuous $\forall x \in \mathbb{R}$ and differentiable $\forall x \in \mathbb{R}$
- B. is continuous $\forall x \in \mathbb{R}$ and differentiable $\forall x \in \mathbb{R}$ except at $x = \frac{3}{2}$
- C. is continuous $\forall x \in \mathbb{R}$ and differentiable $\forall x \in \mathbb{R}$ except at $x = \frac{2}{3}$

D. **is continuous** $\forall x \in \mathbb{R}$ except $x = 3$ and differentiable $\forall x \in \mathbb{R}$

calculus gate2010me engineering-mathematics continuity

Answer key

5.2

Definite Integral (3)

5.2.1 Definite Integral: GATE CSE 2023 | Question: 21



The value of the definite integral

$$\int_{-3}^3 \int_{-2}^2 \int_{-1}^1 (4x^2y - z^3) dz dy dx$$

is _____. (Rounded off to the nearest integer)

gatecse-2023 calculus definite-integral numerical-answers one-mark

Answer key

5.2.2 Definite Integral: GATE CSE 2024 | Set 2 | Question: 6



Let $f(x)$ be a continuous function from $\mathbb{R}$ to $\mathbb{R}$ such that

$$f(x) = 1 - f(2 - x)$$

Which one of the following options is the CORRECT value of $\int_0^2 f(x) dx$?

- A. 0 B. 1 C. 2 D. -1

gatecse2024-set2 calculus definite-integral one-mark

Answer key

5.2.3 Definite Integral: GATE CSE 2025 | Set 2 | Question: 2



The value of x such that $x > 1$, satisfying the equation $\int_1^x t \ln t dt = \frac{1}{4}$ is

- A. $\sqrt{e}$ B. e C. e^2 D. $e - 1$

gatecse2025-set2 calculus definite-integral one-mark

Answer key

5.3

Differentiation (10)

5.3.1 Differentiation: GATE CSE 1996 | Question: 1.6



The formula used to compute an approximation for the second derivative of a function f at a point x_0 is

- A. $\frac{f(x_0 + h) + f(x_0 - h)}{2}$ B. $\frac{f(x_0 + h) - f(x_0 - h)}{2h}$
C. $\frac{f(x_0 + h) + 2f(x_0) + f(x_0 - h)}{h^2}$ D. $\frac{f(x_0 + h) - 2f(x_0) + f(x_0 - h)}{h^2}$

gate1996 calculus differentiation normal

Answer key

5.3.2 Differentiation: GATE CSE 2014 Set 1 | Question: 46



The function $f(x) = x \sin x$ satisfies the following equation:

$$f''(x) + f(x) + t \cos x = 0$$

The value of t is _____.

gatecse-2014-set1 calculus easy numerical-answers differentiation

Answer key

5.3.3 Differentiation: GATE CSE 2014 Set 1 | Question: 6



Let the function

$$f(\theta) = \begin{vmatrix} \sin \theta & \cos \theta & \tan \theta \\ \sin(\frac{\pi}{6}) & \cos(\frac{\pi}{6}) & \tan(\frac{\pi}{6}) \\ \sin(\frac{\pi}{3}) & \cos(\frac{\pi}{3}) & \tan(\frac{\pi}{3}) \end{vmatrix}$$

where

$\theta \in [\frac{\pi}{6}, \frac{\pi}{3}]$ and $f'(\theta)$ denote the derivative of f with respect to θ . Which of the following statements is/are **TRUE**?

- I. There exists $\theta \in (\frac{\pi}{6}, \frac{\pi}{3})$ such that $f'(\theta) = 0$
- II. There exists $\theta \in (\frac{\pi}{6}, \frac{\pi}{3})$ such that $f'(\theta) \neq 0$

- A. I only
- B. II only
- C. Both I and II
- D. Neither I nor II

gatecse-2014-set1 calculus differentiation normal

Answer key

5.3.4 Differentiation: GATE CSE 2016 Set 2 | Question: 02



Let $f(x)$ be a polynomial and $g(x) = f'(x)$ be its derivative. If the degree of $(f(x) + f(-x))$ is 10, then the degree of $(g(x) - g(-x))$ is _____.

gatecse-2016-set2 calculus normal numerical-answers differentiation

Answer key

5.3.5 Differentiation: GATE CSE 2017 Set 2 | Question: 10



If $f(x) = R \sin(\frac{\pi x}{2}) + S$, $f'(\frac{1}{2}) = \sqrt{2}$ and $\int_0^1 f(x) dx = \frac{2R}{\pi}$, then the constants R and S are

- A. $\frac{2}{\pi}$ and $\frac{16}{\pi}$
- B. $\frac{2}{\pi}$ and 0
- C. $\frac{4}{\pi}$ and 0
- D. $\frac{4}{\pi}$ and $\frac{16}{\pi}$

gatecse-2017-set2 engineering-mathematics calculus differentiation

Answer key

5.3.6 Differentiation: GATE CSE 2024 | Set 1 | Question: 1



Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function such that $f(x) = \max\{x, x^3\}$, $x \in \mathbb{R}$, where $\mathbb{R}$ is the set of all real numbers. The set of all points where $f(x)$ is NOT differentiable is

- A. $\{-1, 1, 2\}$
- B. $\{-2, -1, 1\}$
- C. $\{0, 1\}$
- D. $\{-1, 0, 1\}$

gatecse2024-set1 calculus differentiation one-mark

Answer key

5.3.7 Differentiation: GATE CSE 2025 | Set 1 | Question: 21



Consider the given function $f(x)$.

$$f(x) = \begin{cases} ax + b & \text{for } x < 1 \\ x^3 + x^2 + 1 & \text{for } x \geq 1 \end{cases}$$

If the function is differentiable everywhere, the value of b must be _____. (rounded off to one decimal place)

gatecse2025-set1 calculus differentiation numerical-answers one-mark

Answer key

5.3.8 Differentiation: GATE DA 2025 | Question: 14



Consider two functions $f: \mathbb{R} \rightarrow \mathbb{R}$ and $g: \mathbb{R} \rightarrow (1, \infty)$. Both functions are differentiable at a point c . Which of the following functions is/are ALWAYS differentiable at c ? The symbol $\cdot$ denotes product and the symbol $\circ$ denotes composition of functions.

- A. $f \pm g$ B. $f \cdot g$ C. $\frac{f}{g}$ D. $f \circ g + g \circ f$

gateda-2025 calculus differentiation limits multiple-selects one-mark

Answer key

5.3.9 Differentiation: GATE DA 2025 | Question: 4



Let $f(x) = \frac{e^x - e^{-x}}{2}$, $x \in \mathbb{R}$. Let $f^{(k)}(a)$ denote the k^{th} derivative of f evaluated at a . What is the value of $f^{(10)}(0)$? (Note: ! denotes factorial)

- A. 0 B. 1 C. $\frac{1}{10!}$ D. $\frac{2}{10!}$

gateda-2025 calculus differentiation one-mark

Answer key

5.3.10 Differentiation: GATE DA 2025 | Question: 41



Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a twice-differentiable function and suppose its second derivative satisfies $f''(x) > 0$ for all $x \in \mathbb{R}$. Which of the following statements is/are ALWAYS correct?

- A. f has a local minima
 B. There does not exist x and $y, x \neq y$, such that $f'(x) = f'(y) = 0$
 C. f has at most one global minimum
 D. f has at most one local minimum

gateda-2025 calculus maxima-minima differentiation multiple-selects two-marks

Answer key

5.4 Integration (11)

5.4.1 Integration: GATE CSE 1993 | Question: 02.6



The value of the double integral $\int_0^1 \int_0^{\frac{1}{x}} \frac{x}{1+y^2} dx dy$ is _____.

gate1993 calculus integration normal fill-in-the-blanks

Answer key

5.4.2 Integration: GATE CSE 1998 | Question: 8



- a. Find the points of local maxima and minima, if any, of the following function defined in $0 \leq x \leq 6$.

$$x^3 - 6x^2 + 9x + 15$$

- b. Integrate

$$\int_{-\pi}^{\pi} x \cos x dx$$

gate1998 calculus maxima-minima integration normal descriptive

Answer key

5.4.3 Integration: GATE CSE 2000 | Question: 2.3



Let $S = \sum_{i=3}^{100} i \log_2 i$ and $T = \int_2^{100} x \log_2 x dx$.

Which of the following statements is true?

- A. $S > T$
C. $S < T$ and $2S > T$

gatecse-2000 calculus integration normal

Answer key

- B. $S = T$
D. $2S \leq T$

5.4.4 Integration: GATE CSE 2009 | Question: 25



$$\int_0^{\pi/4} (1 - \tan x)/(1 + \tan x) dx$$

- A. 0 B. 1 C. $\ln 2$ D. $1/2 \ln 2$

gatecse-2009 calculus integration normal

Answer key

5.4.5 Integration: GATE CSE 2011 | Question: 31



Given $i = \sqrt{-1}$, what will be the evaluation of the definite integral $\int_0^{\pi/2} \frac{\cos x + i \sin x}{\cos x - i \sin x} dx$?

- A. 0 B. 2 C. $-i$ D. i

gatecse-2011 calculus integration normal

Answer key

5.4.6 Integration: GATE CSE 2014 Set 3 | Question: 47



The value of the integral given below is

$$\int_0^{\pi} x^2 \cos x dx$$

- A. -2π B. π C. $-\pi$ D. 2π

gatecse-2014-set3 calculus limits integration normal

Answer key

5.4.7 Integration: GATE CSE 2014 Set 3 | Question: 6



If $\int_0^{2\pi} |x \sin x| dx = k\pi$, then the value of k is equal to _____.

gatecse-2014-set3 calculus integration limits numerical-answers easy

Answer key

5.4.8 Integration: GATE CSE 2015 Set 1 | Question: 44



Compute the value of:

$$\int_{\frac{1}{\pi}}^{\frac{2}{\pi}} \frac{\cos(1/x)}{x^2} dx$$

gatecse-2015-set1 calculus integration normal numerical-answers

Answer key

5.4.9 Integration: GATE CSE 2015 Set 3 | Question: 45



If for non-zero x , $af(x) + bf(\frac{1}{x}) = \frac{1}{x} - 25$ where $a \neq b$ then $\int_1^2 f(x) dx$ is

A. $\frac{1}{a^2-b^2} \left[a(\ln 2 - 25) + \frac{47b}{2} \right]$
 C. $\frac{1}{a^2-b^2} \left[a(2\ln 2 - 25) + \frac{47b}{2} \right]$

gatecse-2015-set3 calculus integration normal

Answer key

B. $\frac{1}{a^2-b^2} \left[a(2\ln 2 - 25) - \frac{47b}{2} \right]$
 D. $\frac{1}{a^2-b^2} \left[a(\ln 2 - 25) - \frac{47b}{2} \right]$

5.4.10 Integration: GATE CSE 2018 | Question: 16



The value of $\int_0^{\pi/4} x \cos(x^2) dx$ correct to three decimal places (assuming that $\pi = 3.14$) is ____

gatecse-2018 calculus integration normal numerical-answers one-mark

Answer key

5.4.11 Integration: GATE IT 2005 | Question: 35



What is the value of $\int_0^{2\pi} (x - \pi)^2 (\sin x) dx$

- A. -1 B. 0 C. 1 D. π

gateit-2005 calculus integration normal

Answer key

5.5 Limits (15)

5.5.1 Limits: GATE CSE 1993 | Question: 02.1



$\lim_{x \rightarrow 0} \frac{x(e^x - 1) + 2(\cos x - 1)}{x(1 - \cos x)}$ is _____

gate1993 limits calculus normal fill-in-the-blanks

Answer key

5.5.2 Limits: GATE CSE 1995 | Question: 7(B)



Compute without using power series expansion $\lim_{x \rightarrow 0} \frac{\sin x}{x}$.

gate1995 calculus limits numerical-answers

Answer key

5.5.3 Limits: GATE CSE 2008 | Question: 1



$\lim_{x \rightarrow \infty} \frac{x - \sin x}{x + \cos x}$ equals

- A. 1 B. -1 C. ∞ D. $-\infty$

gatecse-2008 calculus limits easy

Answer key

5.5.4 Limits: GATE CSE 2010 | Question: 5



What is the value of $\lim_{n \rightarrow \infty} \left(1 - \frac{1}{n}\right)^{2n}$?

- A. 0 B. e^{-2} C. $e^{-1/2}$ D. 1

gatecse-2010 calculus limits normal

Answer key

5.5.5 Limits: GATE CSE 2015 Set 1 | Question: 4



$\lim_{x \rightarrow \infty} x^{\frac{1}{x}}$ is

- A. ∞ B. 0 C. 1 D. Not defined

Answer key 

5.5.6 Limits: GATE CSE 2015 Set 3 | Question: 9

The value of $\lim_{x \rightarrow \infty} (1 + x^2)^{e^{-x}}$ is

- A. 0 B. $\frac{1}{2}$ C. 1 D. ∞

Answer key 

5.5.7 Limits: GATE CSE 2016 Set 1 | Question: 3



$$\lim_{x \rightarrow 4} \frac{\sin(x-4)}{x-4} = \underline{\hspace{2cm}}$$

Answer key 

5.5.8 Limits: GATE CSE 2017 Set 1 | Question: 28

The value of $\lim_{x \rightarrow 1} \frac{x^7 - 2x^5 + 1}{x^3 - 3x^2 + 2}$

- A. is 0 B. is -1 C. is 1 D. does not exist

Answer key 

5.5.9 Limits: GATE CSE 2019 | Question: 13

Compute $\lim_{x \rightarrow 3} \frac{x^4 - 81}{2x^2 - 5x - 3}$

- A. 1 B. $53/12$
C. $108/7$ D. Limit does not exist

Answer key 

5.5.10 Limits: GATE CSE 2021 Set 1 | Question: 20



Consider the following expression.

$$\lim_{x \rightarrow -3} \frac{\sqrt{2x+22} - 4}{x+3}$$

The value of the above expression (rounded to 2 decimal places) is _____.

Answer key 

5.5.11 Limits: GATE CSE 2022 | Question: 24



The value of the following limit is _____.

$$\lim_{x \rightarrow 0^+} \frac{\sqrt{x}}{1 - e^{2\sqrt{x}}}$$

Answer key 

5.5.12 Limits: GATE DA 2025 | Question: 22



$$\lim_{t \rightarrow +\infty} \sqrt{t^2 + t} - t =$$

(Round off to one decimal place)

gateda-2025 calculus limits numerical-answers one-mark

Answer key

5.5.13 Limits: GATE DA 2025 | Question: 49



Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be such that $|f(x) - f(y)| \leq (x - y)^2$ for all $x, y \in \mathbb{R}$. Then $f(1) - f(0) =$ _____ (Answer in integer)

gateda-2025 calculus limits numerical-answers two-marks

Answer key

5.5.14 Limits: GATE DS&AI 2024 | Question: 50



Evaluate the following limit:

$$\lim_{x \rightarrow 0} \frac{\ln((x^2 + 1) \cos x)}{x^2} =$$

gate-ds-ai-2024 calculus numerical-answers limits engineering-mathematics two-marks

Answer key

5.5.15 Limits: GATE Data Science and Artificial Intelligence 2024 | Sample Paper | Question: 5



$$\lim_{x \rightarrow 2} \frac{\sqrt{x} - \sqrt{2}}{x - 2}$$

- A. 0 B. $\sqrt{2}$ C. $\frac{1}{2\sqrt{2}}$ D. $\frac{1}{\sqrt{2}}$

gateda-sample-paper-2024 calculus limits

Answer key

5.6 Maxima Minima (13)

5.6.1 Maxima Minima: GATE CSE 1987 | Question: 1-xxvi



If $f(x_i) \cdot f(x_{i+1}) < 0$ then

- A. There must be a root of $f(x)$ between x_i and x_{i+1}
 B. There need not be a root of $f(x)$ between x_i and x_{i+1}
 C. The fourth derivative of $f(x)$ with respect to x vanishes at x_i
 D. The fourth derivative of $f(x)$ with respect to x vanishes at x_{i+1}

gate1987 calculus maxima-minima

Answer key

5.6.2 Maxima Minima: GATE CSE 1995 | Question: 1.21



In the interval $[0, \pi]$ the equation $x = \cos x$ has

- A. No solution B. Exactly one solution
 C. Exactly two solutions D. An infinite number of solutions

gate1995 calculus normal maxima-minima

Answer key

5.6.3 Maxima Minima: GATE CSE 1995 | Question: 25a



Find the minimum value of $3 - 4x + 2x^2$.

gate1995 calculus maxima-minima easy descriptive

Answer key

5.6.4 Maxima Minima: GATE CSE 1997 | Question: 4.1



What is the maximum value of the function $f(x) = 2x^2 - 2x + 6$ in the interval $[0, 2]$?

- A. 6 B. 10 C. 12 D. 5.5

gate1997 calculus maxima-minima normal

Answer key

5.6.5 Maxima Minima: GATE CSE 2008 | Question: 25



A point on a curve is said to be an extremum if it is a local minimum or a local maximum. The number of distinct extrema for the curve $3x^4 - 16x^3 + 24x^2 + 37$ is

- A. 0 B. 1 C. 2 D. 3

gatecse-2008 calculus maxima-minima easy

Answer key

5.6.6 Maxima Minima: GATE CSE 2012 | Question: 9



Consider the function $f(x) = \sin(x)$ in the interval $x = [\frac{\pi}{4}, \frac{7\pi}{4}]$. The number and location(s) of the local minima of this function are

- A. One, at $\frac{\pi}{2}$ B. One, at $\frac{3\pi}{2}$
C. Two, at $\frac{\pi}{2}$ and $\frac{3\pi}{2}$ D. Two, at $\frac{\pi}{4}$ and $\frac{3\pi}{2}$

gatecse-2012 calculus maxima-minima normal

Answer key

5.6.7 Maxima Minima: GATE CSE 2015 Set 2 | Question: GA-3



Consider a function $f(x) = 1 - |x|$ on $-1 \leq x \leq 1$. The value of x at which the function attains a maximum, and the maximum value of the function are:

- A. 0, -1 B. -1, 0 C. 0, 1 D. -1, 2

gatecse-2015-set2 set-theory&algebra functions normal maxima-minima

Answer key

5.6.8 Maxima Minima: GATE CSE 2020 | Question: 1



Consider the functions

- I. e^{-x}
II. $x^2 - \sin x$
III. $\sqrt{x^3 + 1}$

Which of the above functions is/are increasing everywhere in $[0, 1]$?

- A. III only B. II only
C. II and III only D. I and III only

gatecse-2020 engineering-mathematics calculus maxima-minima one-mark

Answer key

5.6.9 Maxima Minima: GATE CSE 2023 | Question: 18



Let

$$f(x) = x^3 + 15x^2 - 33x - 36$$

be a real-valued function.

Which of the following statements is/are TRUE?

- A. $f(x)$ does not have a local maximum. B. $f(x)$ has a local maximum.
C. $f(x)$ does not have a local minimum. D. $f(x)$ has a local minimum.

gatecse-2023 calculus maxima-minima multiple-selects one-mark

Answer key

5.6.10 Maxima Minima: GATE DA 2025 | Question: 39



Consider the function $f(x) = \frac{x^3}{3} + \frac{7}{2}x^2 + 10x + \frac{133}{2}$, $x \in [-8, 0]$. Which of the following statements is/are correct?

- A. The maximum value of f is attained at $x = -5$
B. The minimum value of f is attained at $x = -2$
C. The maximum value of f is $\frac{133}{2}$
D. The minimum value of the derivative of f is attained at $x = -\frac{7}{2}$

gateda-2025 calculus maxima-minima multiple-selects two-marks

5.6.11 Maxima Minima: GATE DS&AI 2024 | Question: 40



Consider the function $f: \mathbb{R} \rightarrow \mathbb{R}$ where $\mathbb{R}$ is the set of all real numbers.

$$f(x) = \frac{x^4}{4} - \frac{2x^3}{3} - \frac{3x^2}{2} + 1$$

Which of the following statements is/are TRUE?

- A. $x = 0$ is a local maximum of f B. $x = 3$ is a local minimum of f
C. $x = -1$ is a local maximum of f D. $x = 0$ is a local minimum of f

gate-ds-ai-2024 calculus maxima-minima multiple-selects two-marks

Answer key

5.6.12 Maxima Minima: GATE DS&AI 2024 | Question: 5



For any twice differentiable function $f: \mathbb{R} \rightarrow \mathbb{R}$, if at some $x^* \in \mathbb{R}$, $f'(x^*) = 0$ and $f''(x^*) > 0$, then the function f necessarily has a _____ at $x = x^*$.

Note: $\mathbb{R}$ denotes the set of real numbers.

- A. local minimum B. global minimum
C. local maximum D. global maximum

gate-ds-ai-2024 calculus maxima-minima one-mark

Answer key

5.6.13 Maxima Minima: GATE IT 2008 | Question: 31



If $f(x)$ is defined as follows, what is the minimum value of $f(x)$ for $x \in (0, 2]$?

$$f(x) = \begin{cases} \frac{25}{8x} & \text{when } x \leq \frac{3}{2} \\ x + \frac{1}{x} & \text{otherwise} \end{cases}$$

A. 2

B. $2\frac{1}{12}$

C. $2\frac{1}{6}$

D. $2\frac{1}{2}$

gateit-2008 calculus maxima-minima normal

Answer key

5.7

Polynomials (2)

5.7.1 Polynomials: GATE CSE 1987 | Question: 1-xxii



The equation $7x^7 + 14x^6 + 12x^5 + 3x^4 + 12x^3 + 10x^2 + 5x + 7 = 0$ has

A. All complex roots

B. At least one real root

C. Four pairs of imaginary roots

D. None of the above

gate1987 calculus polynomials

Answer key

5.7.2 Polynomials: GATE CSE 1995 | Question: 2.8



If the cube roots of unity are $1, \omega$ and ω^2 , then the roots of the following equation are

$$(x-1)^3 + 8 = 0$$

A. $-1, 1+2\omega, 1+2\omega^2$

B. $1, 1-2\omega, 1-2\omega^2$

C. $-1, 1-2\omega, 1-2\omega^2$

D. $-1, 1+2\omega, -1+2\omega^2$

gate1995 calculus normal polynomials

Answer key

Answer Keys

5.1.1	N/A	5.1.2	B	5.1.3	A	5.1.4	A	5.1.5	A
5.1.6	C	5.1.7	19 : 19	5.1.8	A	5.1.9	C	5.2.1	0
5.2.2	B	5.2.3	A	5.3.1	D	5.3.2	-2	5.3.3	C
5.3.4	9	5.3.5	C	5.3.6	D	5.3.7	-2.1:-1.9	5.3.8	A;B;C
5.3.9	A	5.3.10	B;C;D	5.4.1	N/A	5.4.2	N/A	5.4.3	A
5.4.4	D	5.4.5	D	5.4.6	A	5.4.7	4	5.4.8	-1
5.4.9	A	5.4.10	0.288 : 0.289	5.4.11	B	5.5.1	1	5.5.2	1
5.5.3	A	5.5.4	B	5.5.5	C	5.5.6	C	5.5.7	1
5.5.8	C	5.5.9	C	5.5.10	0.25 : 0.25	5.5.11	-0.5	5.5.12	0.5:0.5
5.5.13	0:0	5.5.14	0.5	5.5.15	C	5.6.1	A	5.6.2	B
5.6.3	1	5.6.4	B	5.6.5	B	5.6.6	D	5.6.7	C
5.6.8	A	5.6.9	B;D	5.6.10	C;D	5.6.11	A;B	5.6.12	A
5.6.13	B	5.7.1	B	5.7.2	C				